



MEDICINES CONTROL AGENCY
Clinical Trials Unit

WORKSHOP ON PHASE I STUDIES

**ILLUSTRATIVE GUIDANCE FOR APPLICATIONS TO THE COMPETENT
AUTHORITY TO COMMENCE A CLINICAL TRIAL IN EARLY PHASE OF
DEVELOPMENT ENTR/6418/01**

ANNEX 5

ILLUSTRATIVE GUIDANCE FOR APPLICATIONS TO THE COMPETENT AUTHORITY TO COMMENCE A CLINICAL TRIAL IN EARLY PHASE OF DEVELOPMENT ENTR/6418/01

Introduction

This guidance presents a mock application for a fictitious product. The information presented illustrates the quantity and level of detail expected where it is available. If information for a section is not available this should be clearly stated. In the case of products where certain types of data cannot be provided this should be justified briefly. It should be noted assessors/reviewers do not need details of methods for standard tests and prefer data presented as tables with a brief discussion of the results and any conclusion.

The guidance presents illustrative data relating to the pharmaceutical, non-clinical and clinical aspects of an investigational product. It is assumed that applicants will derive this information from tests carried out to the current standards of GMP, GLP and GCP and that they will have followed regulatory guidance on conducting the tests, if available. The application should draw attention to any deviations from these standards and provide a justification. Finally although this illustrative guidance does not provide information under all headings, it is important to note that all available information relating to the investigational product should be provided at the time of the application. Where possible information should be presented using the headings listed in Attachments 1 to 3 of the guideline.

CHEMICAL PHARMACEUTICAL AND BIOLOGICAL DATA

Introduction

This Clinical Trial Application presents information relating to AB1234 Capsules containing 1mg, 2.5mg or 5mg of AB1234 (as the sodium salt). AB1234 is a proton pump inhibitor and is being developed for the treatment of gastro-oesophageal reflux disease.

2.1.S DRUG SUBSTANCE

2.1.S.1 General Information

2.1.S.1.1 Nomenclature

Chemical Name (IUPAC)

The chemical name would be provided

Code Name

AB1234 sodium salt

Other names

2.1.S.1.2 Structure

Structural Formula

The chemical structure would be provided

2.1.S.1.3 General Properties

Molecular Formula

The molecular formula would be presented.

Molecular Weight

The molecular weight would be presented.

Chirality/Stereochemistry

AB1234 is a single stereoisomer with the (2S) configuration.

Description

A yellow crystalline solid.

pH and pKa

pKa of 2.37.

Melting Point

Material melts at approximately 160°C with decomposition at temperatures above approximately 190°C.

Solubility

The solubility of AB1234 sodium salt in water at 20°C is approximately 35 mg/mL. Aqueous solubility is unaffected by pH in the range 1-8.

Hygroscopicity

AB1234 is not considered to be hygroscopic. No increase in moisture content was seen following storage at 25°C/90% RH, for 2 weeks.

Crystal Form

There is only one known crystalline form of AB1234.

2.1.S.2 Manufacture

2.1.S.2.1 Manufacturer(s)

The drug substance, AB1234, is manufactured in accordance with Good Manufacturing Practice at the following facility:

Company Name

Street address

Town

Country

2.1.S.2.2 Description of Manufacturing Process and Process Controls

Synthetic route

XYZ123 was condensed with XYZ456 using thionyl chloride. The resulting intermediate PQR123 was not isolated and was subsequently reacted with XYZ789 to form PQR 456. This was isolated by crystallisation and then dissolved in toluene. Hydrogenation in the presence of a palladium catalyst

resulted in the crude drug substance which was purified by crystallisation from aqueous ethanol.

A flow diagram for the synthesis of AB1234 is provided in Figure 1 including starting materials, intermediates, solvents and reagents (including catalysts) for each stage and indicating stereochemical information where appropriate. Where intermediates are not isolated this is indicated by the use of square brackets.

Synthetic route for batches used in nonclinical studies

Two batches of AB1234 (R1234/01/1 and R5678/01/1) have been used in key nonclinical studies. These batches were manufactured according to the flow diagram provided in Figure 1 but aqueous industrial methylated spirits (IMS) was used as the crystallisation solvent instead of aqueous ethanol.

Figure 1 Flow Diagram for the Synthesis of AB1234

The flow diagram would be presented.

2.1.S.2.3 Control of Materials

Development of the synthesis of the active ingredient is still at an early stage. Appropriate specifications for the starting materials and intermediates are not yet available but will be established as more experience of the manufacturing process is obtained.

[It may be necessary to discuss the stereochemical control of starting materials where chiral compounds are involved.]

2.1.S.2.4 Controls of Critical Steps and Intermediates

No information yet available.

2.1.S.2.5 Process Validation and/or Evaluation

No information yet available.

2.1.S.2.6 Manufacturing Process Development

No information yet available.

2.1.S.3 Characterisation

2.1.S.3.1 Elucidation of Structure and Other Characteristics

The route of synthesis presented in Figure 1 and the spectroscopic studies performed, (Figures 2-6) are consistent with the assigned chemical structure. These data were generated using AB1234 batch R1234/01/1, which was manufactured according to the flow diagram in Figure 1, with the exception that it was recrystallised from aqueous IMS.

The elemental analysis results are in agreement with theoretical values.

Table 1 Elemental Analysis of AB1234 (Batch R1234/01/1)

Molecular Formula		
Element	% Theoretical	% Found
C		
H		
N		

The infrared spectroscopic data are consistent with the chemical structure of AB1234 by assignment of band maxima to the functional groups of the molecule. The IR spectrum is presented in Figure 2.

The signals from a proton nuclear magnetic resonance (¹H-NMR) spectrum were assigned to the protons in the molecule. The signals from a carbon nuclear magnetic resonance (¹³C-NMR) spectrum were assigned to the carbons present in the molecule. All spectra obtained were consistent with

the structure of AB1234. The ^1H -NMR and ^{13}C -NMR spectra and their assignments are presented in Figures 3 and 4 respectively. The electrospray mass spectrum, presented in Figure 5 indicates the spectrum is consistent with the proposed structure. The assignment of ions is presented in Table 3. In 0.1M hydrochloric acid AB123456G exhibits UV absorbance maxima at about 204 nm, 228 nm and 258 nm. The UV spectrum is presented in Figure 6.

Figure 2 Infrared Spectrum of a Mineral Oil Mull of AB1234 (Batch R1234/01/1)

[The IR spectrum would be presented.]

Peak Wavenumber (cm ⁻¹)	Peak Type	Assignment

Figure 3 400 MHz ^1H -NMR Spectrum of AB1234 (Batch R1234/01/1) in 2% w/v DCl in D₂O

[The ^1H -NMR spectrum and a table of assignments would be presented.]

Figure 4 ^{13}C -NMR Spectrum of AB1234 (Batch R1234/01/1) in 2% w/v DCl in D₂O

[The ^{13}C -NMR spectrum and a table of assignments would be presented.]

Figure 5 Electrospray Product Ion Mass Spectrum of AB1234 (Batch R1234/01/1)

[The mass spectrum and a table of assignments would be presented.]

Figure 6 Ultraviolet Absorption Spectrum of AB1234 (Batch R1234/01/1) in 0.1 M HCl

[The UV spectrum and a table of assignments would be presented.]

2.1.S.3.2 Impurities

Potential impurities from the synthesis and degradation of AB1234

Potential drug-related impurities in AB1234 are presented in Table 2.

Other potential impurities which may arise during synthesis are residual palladium catalyst and residual organic solvents (ethanol and toluene) which are controlled by the specification for AB1234.

[The inclusion of a specific test for individual residual solvents may be necessary depending on the solvent(s) used and based on dosing considerations.]

Table 2 Potential Process Impurities and Degradation Products

Code Name	Structural Formula	Notes	Typical Level Observed
		Starting material, reagent, intermediate, degradant, process impurity catalyst etc	

2.1.S.4 Control of Drug Substance

2.1.S.4.1 Specification

Batches of the active ingredient will comply with the following specification. Batches will be released for clinical trial purposes only if the impurity profiles can be supported by available nonclinical data

Table 3 Specification for AB123456G

Test	Acceptance Criteria
Description	Yellow crystalline solid
Identification AB1234 by IR	The spectrum of the sample is concordant with that of the AB1234 authentic material.
AB1234 content by HPLC (% w/w)	Greater than 97
Drug-related impurities content by HPLC (% area) Any unqualified impurity Total impurities	Not greater than 0.5 (<i>limit to be defined based on dosing considerations</i>) Not greater than 3.0
Residual palladium by ICP	Not greater than 3ppm
Loss of Drying (% w/w)	Not greater than 0.5

2.1.S.4.2 Analytical Procedures

The methods used to control the drug substance are summarised below. In the course of ongoing development analytical methods will continue to be optimised and revised methods implemented and appropriately validated.

Description of AB1234

A sample of the drug substance is examined for physical form and colour.

Identification of AB1234by IR

The infrared absorption spectrum of the drug substance is recorded in the range of 4000 cm⁻¹ to 400 cm⁻¹. The spectrum of the sample is compared to the spectrum of an authentic sample to ensure that it is concordant.

AB1234 Content by HPLC

The method for determination of AB1234 content is an isocratic reversed-phase HPLC method, using a 3.5 µm C18 column, or suitably validated alternative. The mobile phase is a mixture of methanol and water, with UV detection at 280 nm.

Drug-related Impurities Content of AB1234 by HPLC

The method for the determination of drug-related impurities content of AB1234 is a reversed-phase gradient HPLC method, using a 5 µm C8 column, or suitably validated alternative. The mobile phase is a mixture of methanol and water, with UV detection at 280 nm.

Loss on Drying of AB123456G

Perform according to USP <731>, drying at an oven temperature of 100°C for 3 hours.

Residual Palladium

The method for the determination of residual palladium is an ICP method.

2.1.S.4.3. Validation of Analytical Procedures

Only brief details of the validation performed are required. There should be sufficient validation data in place to assure that the methods are suitable for use.

AB1234 Content by HPLC

Validation of the method has been carried out to demonstrate specificity, linearity and repeatability.

Drug-related Impurities Content of AB1234 by HPLC

Validation of the method has been carried out to demonstrate specificity, repeatability and the limit of detection for each significant drug-related impurity.

[The LOD should be provided for significant drug-related impurities]

2.1.S.4.4 Batch Analyses

Batch analysis data are presented in Table 4 for one batch of AB1234 manufactured for clinical use using the synthetic route provided in Figure 1, and for two batches of AB1234 manufactured using aqueous IMS and used in the key nonclinical safety studies.

[In addition to reporting impurities content against the specification, further impurity data should be included in the batch analysis table to provided further details of the impurity profile of key batches.]

Table 4 Batch Analysis Data for Batches of AB123456G

Batch Number	Acceptance Criteria	R1234/01/1	R5678/01/1	F00445
Batch Size (kg)		1.15	3.2	5.4
Place of Manufacture		Site specified	Site specified	Site specified
Date of Manufacture		Jun 1999	Dec 2000	July 2001
Use		Nonclinical	Non clinical	Clinical
Test				
Description	Yellow crystalline solid	Yellow crystalline powder	Yellow crystalline powder	Yellow crystalline powder
Identification AB1234 by IR	The spectrum of the sample is concordant with that of the AB1234 authentic material.	Conforms	Conforms	Conforms
AB1234 content by HPLC (% w/w)	Greater than 97	99.2	97.4	99.5
Drug-related impurities content by HPLC, (% area)				
AB2460	For information only	0.14	0.51	0.11
AB125689	For information only	0.23	0.46	<0.05
AB112233	For information only	0.11	0.32	<0.05
Any unqualified impurity	Not greater than 0.X	0.13	0.14	<0.05
Total	Not greater than 3.0	0.74	2.2	0.5
Residual Palladium by ICP	Not greater than 3ppm	1	1	1
Loss of Drying (% w/w)	Not greater than 0.5	<0.1	<0.1	<0.1

2.1.S.5 Reference Standards or Materials

Confirmation of what is used as an authentic sample should be included

2.1.S.6 Container Closure System

AB1234 is stored in double low density polyethylene bags, sealed with plastic ties, and placed in polyethylene (HDPE) kegs.

2.1.S.7 Stability

Decomposition Chemistry

[A brief summary of decomposition studies performed to date would be included.]

Forced degradation studies of AB1234 were conducted on the solid state.

Solid samples of AB1234 (Batch R1234/01/1) were stored at 60°C/75% RH for 6 weeks, at 80°C (under nitrogen, air and oxygen headspaces) for 2 weeks, under fluorescent light (1490 ft. candles) for 8 days.

The principal degradation product was LM1122. The solid state samples showed no significant changes in appearance, AB1234 content, drug-related impurities or physical form under any of the conditions studied.

Stability Studies

[Stability data available to date would be presented].

A stability study is ongoing for one batch of AB1234 manufactured according to the synthetic process described in Section 2.1.S.2.2. and packed in individual sealed polyethylene bags, simulating routine storage of the drug substance. Data are available to date for AB1234 stored for up to 6 months at 30°C/60% RH or 40°C/75% RH, for up to 3 months at 40°C/75% RH Exposed (i.e. in an open polythene bag), up to 3 months at 50°C or for 7 days in a light cabinet. Samples were tested at appropriate intervals for description, AB1234 content and drug-related impurities.

Results and Conclusions

[Depending on the extent of data available, it may be possible to state a retest period. If this is not stated then assurance should be provided on the steps taken to ensure the quality of the drug substance at time of manufacture of the investigational medicinal product]

Results to date are summarised in Table 5. There is no evidence of any significant physical or chemical changes in AB1234 at any of the storage conditions. No significant changes were observed in appearance, AB1234 content or drug-related impurities under any of the storage conditions including light exposure, and all results comply with specification.

AB1234 should be stored at temperatures not exceeding 30°C.

It will be tested to its full specification prior to use in the manufacture of batches of drug product.

Table 5 Summary of Stability Data for AB1234 (Batch R1234/01/1)

Storage Condition	Storage Time (Months)	Mean AB1234 Content by HPLC (% w/w)	Drug-related Impurities by HPLC (% area)			
			LM1122	LM5566	Largest Other Impurity	Total
Initial	0	99.2	0.14, 0.14	0.23, 0.24	0.14, 0.13	0.74, 0.73
30°C/60% RH	3	98.2	0.14	0.20	0.13	0.65
	6	99.1	0.13	0.18	0.12	0.68
40°C/75% RH	3	98.1	0.12	0.15	0.11	0.57
	6	99.5	0.12	0.10	0.11	0.59
40°C/75% RH Exposed	1	99.1	0.17	0.20	0.13	0.77
	3	99.1	0.13	0.14	0.10	0.57
50°C	1	98.9	0.17	0.22	0.14	0.85
	3	98.6	0.13	0.19	0.12	0.66
Light cabinet	7 days	98.9	0.15	0.22	0.12	0.77

2.1.P DRUG PRODUCT

2.1.P.1 Description and Composition of the DRUG Product

Description

AB1234 Capsules, are Size 0 hard gelatin capsule comprising a Swedish orange body and cap and containing 1, 2.5 or 5 mg of AB1234 (present as the sodium salt).

Composition

The complete statement of the components and quantitative composition of AB1234 Capsules is given below in Table 6.

Table 6 Composition of AB1234Capsules

Component	Quantity (mg/capsule)			Function	Reference to Standard
AB1234 sodium salt ¹	1.2	3	6	Active	
Pregelatinized Starch	148.8	147	144	Diluent	PhEur
Sodium Starch Glycolate	25	25	25	Disintegrant	PhEur
Magnesium stearate	5	5	5	Lubricant	PhEur
Total Fill Weight	180	180	180	-	-
Capsule, gelatin	1	1	1	Capsule Shell	Supplier

1. Equivalent to 1, 2.5 or 5 mg AB1234, The actual quantity of AB1234 used may be adjusted based on the purity of each batch of drug substance.

2.1.P.2 Pharmaceutical Development

[Due to the early phase of development there may be no information to include in this section. It may be necessary by exception if discussion of pharmaceutical development, choice of excipients etc is appropriate to aid understanding of a novel or complex formulation.]

2.1.P.3 Manufacture

2.1.P.3.1 Manufacturer(s)

[The site of manufacture only may be stated but sites of assembly would be required to be included in the application form].

The manufacture of AB1234 Capsules is conducted in accordance with Good Manufacturing Practice at the following facilities:

Company Name
Street address
Town
Country

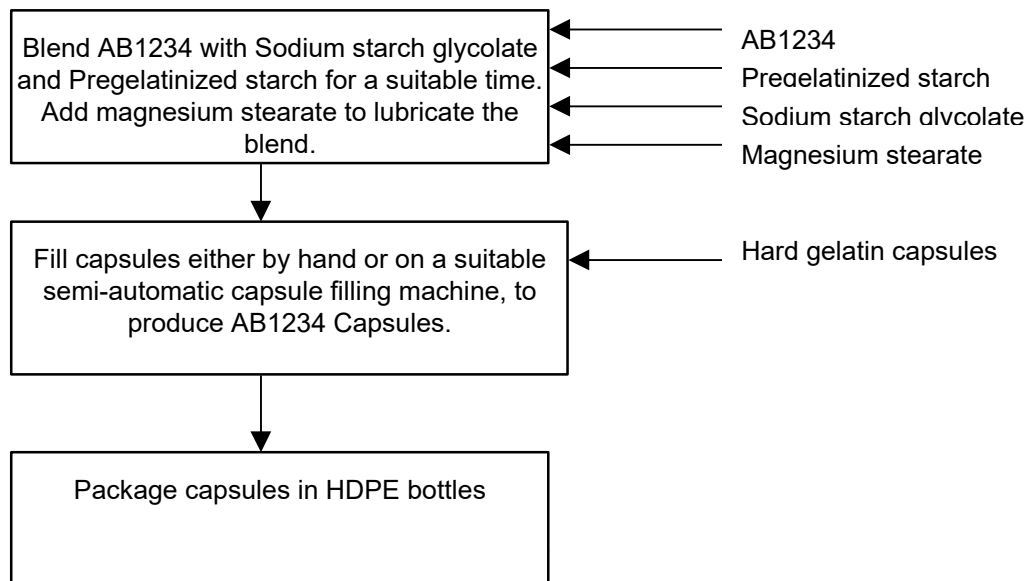
2.P.3.2 Batch Formula

For simple dosage forms it may not be necessary to include this information in the CTA at this stage of development.

2.1.P.3.3 Description of Manufacturing Process and Process Controls

The flow diagram of the manufacturing process of AB1234 Capsules is given in Figure 7.

Figure 7 Flow Diagram of the Manufacturing Process of AB1234 Capsules 1, 2.5 and 5 mg



In-process Controls

[It would not be anticipated that in-process controls would be specified at this stage of development with the exception of those for parenteral products or where they are required to ensure final product quality i.e. fill weight for powder in bottle formulations. If the section is not completed it would be deleted.]

2.1.P.3.4 Controls of Critical Steps and Intermediates

[Information will not usually be required in early stage development but may be necessary for complex or novel products.]

[With the exception of parenterals this section would not be not be presented. If the section is not completed it would be deleted.]

2.1.P.3.5 Process Validation and/or Evaluation

[Information will not usually be required in early stage development but may be necessary where this impacts on the safety of the product.]

2.1.P.4 Control of Excipients

2.1.P.4.1 Specifications

The following excipients used in the manufacture of the finished product are required to comply with the specifications listed below.

These excipients will be tested for compliance with specifications until such time as confidence is gained in respect of approved suppliers, when reduced testing may be implemented. In such cases, receipt of a Certificate of Analysis from the supplier will be required which shows full compliance with specification. Additionally, sufficient identity tests will be implemented to confirm the identity of the material.

Pregelatinized starch, sodium starch glycolate and magnesium stearate comply with the requirements of the Ph.Eur. The gelatine capsules are non-pharmacopoeial but the gelatine complies with the requirements of the Ph. Eur. The magnesium stearate is of vegetable origin.

2.1.P.4.5 Excipients of Human or Animal Origin

The gelatine in the capsules is of bovine origin. An EDQM Certificate of Suitability (RO-CEP 2000-xxx Rev 00 dated aa.bb.cccc).

2.1.P.4.6 Novel Excipients

None involved.

[For novel excipients further information to demonstrate the safety of the material would be provided. If the section is not completed it would be deleted]

2.1.P.5 Control of DRUG Product

2.1.P.5.1 Specification(s)

Clinical trial batches will meet the following specification at release.

Table 7 Regulatory Specification for AB1234 Capsules 1, 2.5 and 5 mg

Test	Acceptance Criteria
Description	A white to off-white powder contained in a Size 0 hard gelatin capsule comprised of a Swedish orange body and cap

Identification of AB1234 by HPLC	The retention time of the principal peak in the sample chromatogram corresponds to that of the principal peak in the standard chromatogram
AB1234 content by HPLC	90.0 – 110.0% of label claim
Degradation products by HPLC	Report results
Dissolution (% AB1234 released)	Complies with USP, Q = 75 at 45 minutes

2.1.P.5.2 Analytical Procedures

The methods used to control the drug substance are summarised below. In the course of ongoing development, analytical methods will continue to be optimised and revised methods implemented and appropriately validated.

Description of AB1234 Capsules

The capsules are examined for conformance of colour and shape. The contents are tipped out and examined against a white background

Identification and Content of AB1234 in AB1234 Capsules by HPLC

This method uses a LiChrosphere 100 RP18, 5µm, 250 x 4mm column equilibrated at 40°C. The eluent is a linear gradient starting with 70% water with 0.05% trifluoroacetic acid and 30% acetonitrile with 0.05% trifluoroacetic acid and changing to 30% water with 0.05% trifluoroacetic acid and 70% acetonitrile with 0.05% trifluoroacetic acid over 20 minutes. Detection is by UV at a wavelength of 274 nm. This method is stability indicating and will detect any degradation products present.

Dissolution of AB1234 Capsules

The dissolution of AB1234 Capsules is determined using Apparatus II of the USP, with a dissolution medium of 900 mL of ammonium acetate buffer at pH 4.0 and a paddle speed of 50 rpm. Detection is by UV absorbance at 262 nm.

2.1.P.5.3. Validation of Analytical Procedures

At this early phase only brief details of the validation performed would be provided, but sufficient validation data would be in place to assure that the methods are suitable for use.

Identification and Content of AB1234 in AB1234 Capsules by HPLC

The method has been validated for specificity, linearity and repeatability.

Dissolution of AB1234 Capsules

The method has been validated for specificity, linearity and repeatability.

2.1.P.5.4 Batch Analyses

Batch details and batch analysis data are provided in Table 8 for one batch of each strength of AB1234 Capsules. All batches were manufactured according

to the details described in Section 2.1.P.5.1 and tested by the methods described in Section 2.1.P.5.2.

2.1.P.5.5 Characterisation of Impurities

[Include any information available to date.]

[It would be proposed that this section is used by exception dependent on the product. Where it is needed, a brief textual summary would be included of the critical impurities specific to the formulation, based on safety considerations. If the section is not completed it would be deleted.]

2.1.P.5.6 Justification of Specification(s)

Batches will be released for clinical trial purposes only if the impurity profiles can be supported by available nonclinical data

Table 10 Batch Analysis Data for Batches of AB1234 Capsules, 1 mg

Batch Number		1111
Input Drug Substance Batch Number		F004495
Batch Size (Units)		252
Date of Manufacture		May 2002
Site of Manufacture		Company Name, Town
Use		Clinical
Test	Acceptance Criteria	
Description	A white to off-white powder contained in a size 0 hard gelatin capsule comprised of a Swedish orange body and cap	Conforms
Identification	The retention time of the principal peak in the sample chromatogram corresponds to that of the principal peak in the standard chromatogram	Conforms
AB123456 Content	0.9 to 1.1 mg per capsule (90.0 – 110.0% of label claim)	1.02 mg (102.0 % of label claim)
Degradation Products	Report result	None detected
Dissolution	Complies with USP, Q = 75% at 45 minutes	99 % at 45 minutes

Batch Analysis Data for Batches of AB1234 Capsules, 2.5 mg

Batch Number		2222
Input Drug Substance Batch Number		F004495
Batch Size (Units)		290
Date of Manufacture		May 2002
Site of Manufacture		Company Name, Town
Use		Clinical
Test	Acceptance Criteria	
Description	A white to off-white powder contained in a size 0 hard gelatin capsule comprised of a Swedish orange body and cap	Conforms
Identification	The retention time of the principal peak in the sample chromatogram corresponds to that of the principal peak in the standard chromatogram	Conforms
AB123456 Content	2.25 to 2.75 mg per capsule (90.0 – 110.0% of label claim)	2.45 mg (98.0 % of label claim)
Degradation Products	Report result	None detected
Dissolution	Complies with USP, Q = 75% at 45 minutes	97 % at 45 minutes

Batch Analysis Data for Batches of AB1234 Capsules, 5 mg

Batch Number		3333
Input Drug Substance Batch Number		F004495
Batch Size (Units)		220
Date of Manufacture		May 2002
Site of Manufacture		Company Name, Town
Use		Clinical
Test	Acceptance Criteria	
Description	A white to off-white powder contained in a size 0 hard gelatin capsule comprised of a Swedish orange body and cap	Conforms
Identification	The retention time of the principal peak in the sample chromatogram corresponds to that of the principal peak in the standard chromatogram	Conforms
AB123456 Content	4.5 to 5.5 mg per capsule (90.0 – 110.0% of label claim)	5.1 mg (102.0 % of label claim)
Degradation Products	Report result	None detected
Dissolution	Complies with USP, Q = 75% at 45 minutes	98 % at 45 minutes

2.1.P.6 Reference Standards or Materials

[Due to the early phase of development it would not be proposed that any information on reference standards or materials would be presented and the section would be deleted.]

2.1.P.7 Container Closure System

The capsules are packaged into HDPE bottles, and closed with child-resistant plastic induction seal screw caps.

2.1.P.8 Stability

Stability studies are being conducted with testing for content, dissolution and degradation products at regular intervals to confirm the stability of AB1234 Capsules over the proposed study duration. Samples of the 1mg and 5mg capsules have been stored at 25°C/60%RH and at 40°C/75%RH for 6 weeks. The results are provided in Table 11.

AB123456 Capsules will be stored at temperatures not exceeding 30°C with a shelf life of 6 months.

Table11 Summary of Stability Data for AB1234 Capsules 1mg

Storage Condition	Storage Time (Weeks)	Mean AB1234 Content by HPLC (% w/w)	Dissolution (% at 45 minutes)	Degradation Products by HPLC (% area)			
				LM1122	LM5566	Largest Other Impurity	Total
Initial	0	102.0	99.0	None detected	None detected	None detected	None detected
25°C/60% RH	2	101.2	97.0	None detected	None detected	None detected	None detected
	6	99.1	100.0	0.10	None detected	None detected	0.10
40°C/75% RH	2	98.1	98.0	0.12	< 0.10	None detected	0.15
	6	99.5	96.0	0.20	0.10	None detected	0.30

Summary of Stability Data for AB1234 Capsules 5mg

Storage Condition	Storage Time (Weeks)	Mean AB1234 Content by HPLC (% w/w)	Dissolution (% at 45 minutes)	Degradation Products by HPLC (% area)		
				LM1122	LM5566	Largest Other Impurity
Initial	0	102.0	98.0	None detected	None detected	None detected
25°C/60% RH	2	103.0	99.0	None detected	None detected	None detected
	6	99.1	96.5	0.23	None detected	None detected
40°C/75% RH	2	99.1	98.0	0.19	0.11	None detected
	6	97.5	92.0	0.25	0.15	None detected

2.1.AAPPENDICES

2.A.1 Facilities and Equipment

Applicable to biotech products only and the section would be deleted.

2.1.A.2 Adventitious Agents Safety Evaluation

Gelatine used in the gelatine capsules has an EDQM Certificate of Suitability (RO-CEP 2000-xxx Rev 00 dated aa.bb.cccc).

[Information would be presented if appropriate. If the section is not completed it would be deleted.]

2.1.A.3 Novel Excipients

None involved.

[Information would be presented if appropriate. If the section is not completed it would be deleted.]

2.1.A.4 Solvents for Reconstitution and Diluents

None involved.

PART III PHARMACO-TOXICOLOGICAL DATA

INTRODUCTION

This is a randomised, placebo controlled, study in approximately 20 healthy male volunteers. AB 1234 reduces gastric acid secretion by inhibition of the gastric proton pump in the parietal cell. It is rapidly acting and animal data provides evidence of reversible control of gastric acid secretion with once daily dosing.

The batches of compound used in the toxicology studies had a similar impurity profile to those to be used in the proposed clinical trial.

PHARMACOLOGY

Summary

AB1234 has been shown to be a selective proton pump inhibitor. The proton pump inhibitory activity has been demonstrated in *in vitro* functional studies and its acid reducing activity has been confirmed in a number of *in vivo* animal models. No significant activity has been demonstrated at 32 other receptor types investigated, with the exception that at high concentrations there was some evidence of α_1 and α_2 -adrenoceptor binding activity.

Primary Pharmacodynamics

AB1234 was shown to be 10 times more potent than omeprazole in the inhibition of pig H^+/K^+ -ATPase with IC_{50} values of $2.0 \times 10^{-7}M$ and $2.6 \times 10^{-6}M$ respectively. It showed approximately 5% of the activity of ouabain in inhibiting Na^+/K^+ -ATPase. The IC_{50} value for the inhibition of stimulated acid secretion in isolated rabbit glands was $1.9 \times 10^{-7}M$. Reversibility of inhibition was observed in studies with pig gastric glands.

The acid secretion effect of single intraduodenal doses of 1,3,10 or 30 $mg \cdot kg^{-1}$ AB1234 on acid output, pepsin secretion and/or total volume were studied in rats with a ligated pylorus (Shay model) in which endogenous parasympathetic mediators stimulate acid secretion and in rats subjected to exogenous stimulation (subcutaneous injection of histamine). Acid output was inhibited to 10% of control values at the 30 $mg \cdot kg^{-1}$ dose. The inhibition of acid secretion was a dose-dependent effect. No significant effect was seen on pepsin secretion or total volume.

In the dog chronic gastric fistula model with acid secretion stimulated by either histamine or pentagastrin, single and repeat daily intraduodenal doses inhibited acid secretion with an ED_{50} of 0.06 $mg \cdot kg^{-1}$.

In the 'Shay' model, no ulcers were seen in animals receiving AB1234 10mg·kg⁻¹.

The effects of AB1234 on gastric acid secretion and histamine metabolism in the gastric mucosa in two strains of rat were examined. Urinary excretion of gastrin and histamine were increased following the administration of AB1234 10mg·kg⁻¹.

Secondary Pharmacodynamics

[Include brief details, Do not include extensive information on study designs]

SAFETY PHARMACOLOGY

Summary

Safety Pharmacology studies were completed in accordance to the relevant ICH/CPMP Note for Guidance. At high doses AB1234 produces transient hypotension, tachycardia, and slight increase in respiration following intravenous administration but not following oral dosing. CNS effects were only seen at near lethal doses. GI tract effects were those anticipated from the pharmacology of the product.

[Tabulated results helpful. Extensive information on study design not needed. Include details of all doses and routes used. Provide justification if ICH/CPMP guidance document not followed.]

Test	Species	Dose	Results
Gastrointestinal function	rat/mouse	50 mg·kg ⁻¹ iv	As expected from pharmacology
Respiratory function	dog	10 mg·kg ⁻¹ iv	Slight increase in respiration
Renal function	dog	10 mg·kg ⁻¹ iv	Transient decrease in blood flow
Cardiovascular system	Conscious dog	10 mg·kg ⁻¹ iv	Decreased blood pressure Tachycardia
Behaviour/CNS activity	mouse	30 mg·kg ⁻¹ iv 100 mg·kg ⁻¹ iv	No adverse findings Convulsions in 1/6 animals

[Include any data on Toxicokinetics to enable discussion of pharmacokinetics/ pharmacodynamic relationships.]

PHARMACOKINETICS

Summary

Pharmacokinetic data were generated in the mouse, rat and dog . Kinetics would appear to be linear in all species at the doses tested. There were no apparent sex differences in the ADME profile of AB1234.

[Avoid discussing species separately and instead discuss absorption, then distribution, metabolism and excretion. If data generated as part of toxicology study (i.e. Toxicokinetic data) include here and not as separate section.]

AB1234 was rapidly absorbed following oral dosing ($T_{\max} = 0.3\text{-}0.75\text{ h}$) in all species studies (mouse, rat and dog). Oral absolute bioavailability was 43% in the mouse and rat and 66% in the dog.

The volume of distribution was high in all species, indicating that AB1234 is extensively distributed. Whole body radiography studies in the rat indicate that the radiolabelled material was rapidly distributed to all organs with very limited CNS penetration. Organs that showed the highest levels of radiolabelled material were the excretory organs and the adrenals.

In vitro studies in isolated hepatocytes from the different species indicate that there are two major metabolites (M1 and M2). There were no apparent qualitative differences between animals and man. There was evidence of enzyme induction following repeat dosing with doses greater than $150\text{ mg}\cdot\text{kg}^{-1}\cdot\text{day}^{-1}$ in rats as evidenced by a smaller AUC value after 14 days treatment at this dose compared to Day 1. *In vitro* data to date would indicate that AB1234 is primarily metabolised by CYP3A4.

In vivo studies showed evidence of enzyme induction following repeat dosing with doses greater than $150\text{ mg}\cdot\text{kg}^{-1}\cdot\text{day}^{-1}$ in rats as evidenced by a smaller AUC value after 14 days treatment at this dose compared to Day 1.

Excretion studies would indicate that AB1234 is excreted predominately in the urine of the mouse (90% of dose), but urinary and faecal elimination were equally important in other species. There was significant enterohepatic circulation of AB1234 in the rat.

No pharmacodynamic or kinetic drug interaction studies have been performed.

[A table like the following is extremely useful. Inclusion of anticipated HUMAN plasma concentration believed to be required to produce meaningful clinical effect very helpful.]

Summary of kinetic data for AB1234 in animal species following oral administration.

Species	Dose ($\text{mg}\cdot\text{kg}^{-1}\cdot\text{day}^{-1}$)	$C_{\max}$ ($\text{ng}\cdot\text{mL}^{-1}$)	$T_{\max}$ (hrs)	AUC ₍₀₋₂₄₎ ($\text{ng}\cdot\text{h}/\text{ml}$)	$T_{1/2}$ (h)
Mouse	100	300	0.3	432	0.5
Rat	10	30	0.5	152	1

	75 150	145 460	0.5 0.7	1100 2300	1 1.25
Dog	10 30 90	40 125 350	0.75 0.75 0.75	201 612 1750	4 4 4.2

TOXICOLOGY

Summary

All studies were performed in accordance with Good laboratory practice (GLP). Repeat dose studies of up to 14 days duration were performed in the rat and dog as metabolic data indicate that these are relevant species. The duration of the studies is sufficient to allow the proposed trial to commence.

[Tabulated data is useful. There is no need to provide extensive details of study designs. Include details of ongoing studies].

Species	Dose (mg·kg ⁻¹ ·day ⁻¹)	Route	Duration	Findings
Single Dose Studies				
Mouse	200 – 2000	p.o.	14 days observation.	Maximum non-lethal dose 800 mg·kg ⁻¹ , no specific target organ toxicity.
Rat	200 – 2000	p.o.	14 day obs.	Maximum non-lethal dose 850 mg·kg ⁻¹ , no specific target organ toxicity.
Repeat dose studies				
Rat	0, 20, 60, 150	p.o.	14 days	Dose-related increase in plasma gastrin levels, Increase in stomach weights (≥ 60 mg·kg ⁻¹ ·day ⁻¹) with increased thickening of the mucosa. Similar effects have been seen with other 'azoles'. Slight hepatocellular hypertrophy at 150 mg·kg ⁻¹ ·day ⁻¹ .
Dog	0, 10, 30, 90	p.o.	14 days	Dose-related increase in plasma gastrin levels, Increase in stomach weights (90 mg·kg ⁻¹ ·day ⁻¹) with increased thickening of the mucosa. Similar effects have been seen with other 'azoles'.
Rat	0, 10, 50, 100	p.o.	3 month	Ongoing
Dog	0, 10, 30,	p.o.	3 month	Ongoing

	90			
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Reproductive Toxicology

No studies have been completed to date, preliminary teratology studies are ongoing. Examination of testes and sperm parameters on rat repeated dose toxicity studies revealed no adverse effects.

Mutagenicity and Genotoxicity

Ames	200 – 2500 $\mu\text{g}\cdot\text{plate}^{-1}$	Negative, $\pm$ S9 mix, evidence of toxicity at the top concentrations.
Human lymphocyte, CA test	200 – 1750 $\mu\text{g}\cdot\text{mL}^{-1}$	Negative, $\pm$ S9 mix, evidence of toxicity at the top concentrations.

Discussion and Conclusion

The pharmacological profile of AB1234 is considered to support the proposed clinical trial(s). The pharmacokinetics are uncomplicated with the possible exception of enzyme induction at high doses. This enzyme induction was probably responsible for the hepatocellular hypertrophy and should not be problematic in man. The only significant finding in the animal studies was a dose-related increase in plasma gastrin levels, which is an anticipated effect seen with proton pump inhibitors. No other target organ toxicity was observed in rats and dogs respectively.

[Include any recommendations for specific monitoring to be used on proposed trials. Include recommendations for maximum dose/plasma concentrations].

[Tabulated summaries of pivotal studies (in the new Common Technical Document format) may be included as an appendix if it is believed these will be helpful to the reviewer/assessor.]

CLINICAL DATA

Introduction

The proposed clinical trial is a single dose, open label, dose escalation pharmacokinetic/ pharmacodynamic study in healthy male volunteers aged 18 to 45 years. The starting dose is x mg and this will be increased in steps of y mg according to a predetermined protocol.

Study Design

The study design is an open label, ascending dose study with each dose administered to cohorts of 3 volunteers.

Starting dose and dose escalation

The starting dose will be x mg. This has been based on the preclinical pharmacokinetic data generated. A summary of the key data supporting the choice of starting dose has been provided in Table 1. The dosing increments used in the dose escalation will be y mg.

Table 1.

[Provide a summary of the key data supporting choice of dose and dosing increments.]

Endpoints Studied

[Provide a summary of the endpoints to be studied, including the timepoints for blood sampling for the PK/PD endpoints.]

Safety Monitoring

The safety monitoring to be undertaken is provided in Table 2. Since no safety signals were observed in the animal studies, this has been restricted to the standard battery of haematological and biochemistry parameters.

Table 2.

[Provide a flowchart listing the safety parameters to be measured along with the timepoints at which these are considered.]